Project Plan — Monitoring and Predicting Droughts on National Scale: A Satellite-Integrated, Groundwater-Coupled Hybrid Modelling Framework for Denmark

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Abstract

This project plan outlines a three-year PhD research programme integrating satellite data assimilation, groundwater coupling, and hybrid physics-machine learning modelling for operational drought monitoring and forecasting at national scale in Denmark.

Keywords: *drought monitoring; data assimilation; VIC model; groundwater coupling; GRACE-FO; differentiable hydrology; Denmark*

1. PROBLEM STATEMENT AND MOTIVATION

The 2018 Danish drought caused agricultural losses exceeding DKK 6 billion and exposed a structural gap in the national early warning landscape. Operational systems such as **HIP** and the **DK model** deliver high resolution diagnostics but lack probabilistic, satellite integrated outlooks at sub-seasonal to seasonal lead times. Meanwhile, macroscale land surface models such as **VIC**, the model targeted by the DREAM call, lack a prognostic groundwater state. In a country where more than 99% of drinking water originates from aquifers, where water tables typically lie between 1 and 5 metres below surface, and where more than half of the agricultural area is sub-surface drained, this gap is not a detail but a first-order limitation. The recent finding by Forootan *et al.* (2024, *STOTEN* 912:169476) that the severity of the 2017 to 2021 European drought is systematically underestimated in non-assimilated **W3RA** confirms that closing this loop, through groundwater coupling, multi-source data assimilation, and hybrid physics and machine learning integration, is both timely and necessary. The PhD will deliver a satellite-fed VIC Denmark system producing a probabilistic four-pillar combined drought index and short- to seasonal-scale predictions of water dynamics.

2. RESEARCH QUESTIONS

RQ1. Can a groundwater-coupled VIC Denmark, calibrated and constrained by joint assimilation of **GRACE FO** terrestrial water storage, **SMAP** and **ESA CCI** soil moisture, streamflow, and Jupiter piezometric heads, reproduce the propagation of meteorological, agricultural, hydrological, and groundwater drought signals at national scale better than uncoupled or non-assimilated baselines?

RQ2. Does a hybrid framework, combining differentiable parameter learning for VIC parameters with **LSTM** or Transformer post-processing trained on **CAMELS DK**, recover predictive skill in groundwater-dominated Danish catchments, where pure LSTMs underperform (Liu *et al.* 2024)?

RQ3. Can a probabilistic four-pillar combined index (**SPI**/**SPEI** plus **SSMI** plus **SSI** plus **SGI**), aggregated via copulas with full ensemble and posterior uncertainty propagation, classify drought severity more skilfully than **EDO CDI v4**, and improve seasonal predictability when forced with bias-corrected **SEAS5** and **AIFS ENS** or GenCast?

3. APPROACH AND WORK PACKAGES

The project is organised into five interconnected work packages following a de-risked sequence. A physics-based backbone is established in Year 1, and hybrid machine learning extensions are layered on top in Years 2 and 3.

WP1. VIC Denmark with Groundwater Coupling. I will build a national VIC 5 implementation at 1 to 10 km, parameterised from **VIC-Global** and refined with Danish hydrostratigraphy from **GEUS** and Jupiter wells. The groundwater module will start from a lumped per-cell aquifer, following the philosophy of **WGHM** and **SIMGM** (Niu *et al.* 2007), with a clear escalation path to 2D lateral coupling (Zhang and Liang 2021) only if validation justifies the added complexity. This staged design directly addresses the most acute structural limitation of VIC, the leaky bucket lower boundary, without committing prematurely to a fully distributed **MODFLOW**-style solver.

WP2. Joint Calibration and Multi-Source Data Assimilation. I will implement augmented state ensemble Kalman filtering (**LETKF**) and Constrained Bayesian **MCMC** (**ConBay**) for simultaneous parameter calibration and state estimation. Observation operators will be built for **GRACE FO** terrestrial water storage (vertical column sum, sub-monthly L1B following Retegui Schiettekatte *et al.* 2025), **SMAP** L4 root zone and **ESA CCI** surface soil moisture (**CDF** matching, De Lannoy and Reichle 2016), routed streamflow (Lohmann routing or CaMa Flood), and groundwater heads (via effective porosity). The framework will build directly on the group's **PyGLDA** infrastructure (Yang *et al.* 2025, *GMD* 18:6195), addressing the Denmark-to-GRACE scale mismatch through sub-monthly assimilation, ConBay merging with SMAP (Mehrnegar *et al.* 2023), and proper Fennoscandian **GIA** correction (**ICE-6G_D**).

WP3. Hybrid Extension via Differentiable Hydrology. In Year 2 I will train a differentiable VIC variant, following the dPL approach of Tsai *et al.* (2021) and Feng *et al.* (2022), on **CAMELS DK** (Liu *et al.* 2025, *ESSD* 17:1551), Jupiter piezometric records, and satellite forcing. This replaces per-basin **SCE-UA** calibration with end-to-end gradient-based parameter learning across all 3,330 basins, including ungauged ones. An **LSTM** or Transformer residual post-processor (**NeuralHydrology HybridModel**) will correct systematic biases, directly addressing the Liu *et al.* (2024, *HESS* 28:2871) finding that pure LSTMs fail in groundwater-dependent Danish catchments.

WP4. Probabilistic Four-Pillar Combined Drought Index. For each grid cell and time step I will compute standardised sub-indices span-

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ning the full propagation chain: atmospheric (**SPI** or **SPEI** from VIC forcing), agricultural (**SSMI** from root zone soil moisture constrained by **SMAP** and **ESA CCI**), surface hydrological (**SSI** or **SRI** from routed runoff constrained by streamflow observations), and groundwater (**SGI** from the new VIC groundwater state constrained by **GRACE FO** and Jupiter). Sub-indices will be aggregated via Gaussian or vine copulas, producing a probabilistic **MCDI** for each cell and time step with full posterior probabilities of falling within **USDM** or **CDI** severity categories. To my knowledge, a probabilistic, four-pillar, satellite data-assimilation-driven, copula-aggregated drought index with an explicit groundwater pillar has not yet been produced for Denmark. This is the unifying scientific deliverable of the thesis.

WP5. Short- to Seasonal-Scale Drought Forecasting. I will force the calibrated VIC DK with bias-corrected **SEAS5** and **AIFS ENS** or GenCast (Price *et al.* 2025) for 1 to 6 month outlooks. Evaluation will cover hindcasts from 1993 to present and the 2018, 2022, and 2023 events, using **CRPS**, **Brier score**, and **ROC**, with reverse **ESP** diagnostics to disentangle initial-condition skill from forcing skill. Benchmarks include **EFAS Seasonal** and **EDgE**.

4. EXPECTED CHALLENGES AND MITIGATION STRATEGIES

I anticipate four principal challenges.

Challenge 1: Denmark-to-GRACE Scale Mismatch. The native resolution of **GRACE** is around 300 km versus a national area of about 43,000 km^2. This will be mitigated through sub-monthly data assimilation, ConBay merging with higher-resolution **SMAP**, machine learning-based downscaling validated against Jupiter, and rigorous **GIA** correction.

Challenge 2: Equifinality and Identifiability. Joint state and parameter estimation under large observation uncertainty risks multiple parameter sets fitting equally well. This will be mitigated through **GEUS**-derived priors on aquifer parameters, localisation and inflation in **LETKF**, and dPL as a deterministic precursor before stochastic data assimilation.

Challenge 3: Integrating Machine Learning Without Losing Physical Consistency. Adding neural components can erode interpretability and physical fidelity. This will be handled by keeping VIC DK as the physical backbone, restricting the **LSTM** to a residual post-processor role, and applying conformal prediction for honest uncertainty bounds.

Challenge 4: Combined Index Collapse into Smoothed Average. A multi-pillar index risks collapsing into a smoothed average, losing signal contrast. This will be mitigated by pillar-appropriate time scales (**SPI** on 1–3 months, **SGI** on 6–12 months), copula aggregation that preserves tail dependence, and hierarchical validation against Naturskaderådet impact claims, crop yield anomalies, and **EDO CDI v4**.

5. FEASIBILITY, FIT AND OUTPUTS

The plan is feasible within three years because Year 1 already delivers a publishable physics-based baseline, even if the hybrid extensions slip. The **AAU PyGLDA**, ConBay, and **HPC Claaudia** infrastructure provide a strong starting point, while **CAMELS DK**, Jupiter, **DMI Klimagrid**, and **HIPRAD** supply the data backbone. The project aligns with the Villum Young Investigator Programme on "Synergy of Physics-based and Data-driven Methods" and complements **GEUS DK model** expertise, opening natural collaboration with Henriksen, Stisen, and Schneider.

Expected outputs include:

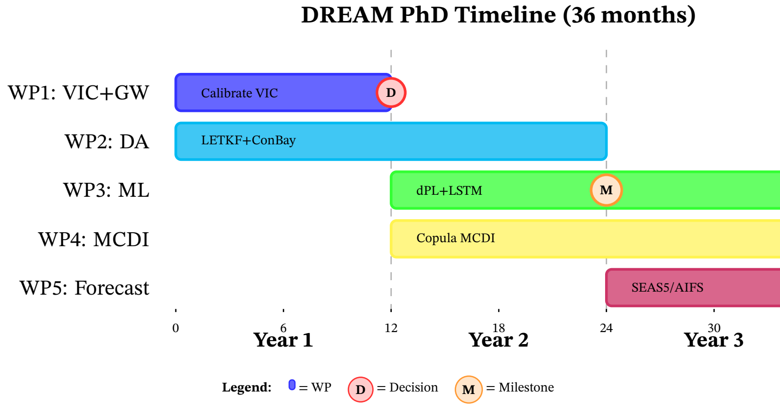

Figure 1. DREAM PhD timeline showing five interconnected work packages over 36 months. **WP1** establishes a physics-based VIC-Denmark baseline with groundwater coupling (Year 1), followed by a critical decision (D) on 2D lateral coupling. **WP2** (LETKF + ConBay data assimilation) runs in parallel throughout, constraining all physical parameters. **WP3** (differentiable hydrology + LSTM hybrid) and **WP4** (copula-aggregated multi-pillar drought index) execute in Years 2–3 in parallel. **WP5** (SEAS5/AIFS seasonal forecasting) focuses on Year 3 outputs. The timeline demonstrates de-risking: Year 1 alone delivers a publishable physics baseline even if hybrid extensions slip. Milestones (M) mark completion of hybrid competence (month 24) and operational prototype readiness (month 36).

- Three first-author papers in *Water Resources Research*, *Hydrology and Earth System Sciences*, and *Remote Sensing of Environment* or *Science of The Total Environment*
- An open-source **VIC DK** plus ConBay data assimilation codebase released through PyGLDA
- An operational **MCDI** prototype co-designed with **DMI**, **GEUS**, and Naturskaderådet end-users

6. BACKGROUND AND ADDED VALUE

My **MSc** thesis at Politecnico di Milano, where I built, calibrated, and ran sensitivity and uncertainty analysis on the **FEST** process-based hydrological model over the Po basin, gave me direct experience of the central methodological tension that this project addresses: constraining subsurface parameters (hydraulic conductivity, deep percolation factors, **SCS** curve number) from surface observations alone.

My current operational role at Progesi S.p.A., where I run multi-hazard monitoring with **ICON** and **ECMWF IFS** forcing, satellite-derived diagnostics, and Python, R, and MATLAB workflows for Regione Lombardia, translates directly to the operational demands of a national-scale drought early warning system.

The combination of process-based modelling, applied data science, and recent exposure to operational hydrometeorology positions me to start contributing to WP1 and WP2 immediately, while developing the hybrid machine learning competence required for WP3 through structured upskilling in **NeuralHydrology**, PyTorch, and differentiable hydrology.

APPENDIX: GLOSSARY OF ACRONYMS

> **Note**
>
> The following acronyms and abbreviations are used throughout this project plan. Definitions are provided at first occurrence in the text; this glossary serves as a reference guide.

- **AIFS ENS** — Artificial Intelligence Foundation Series Ensemble (ECMWF)
- **AAU** — Aalborg University
- **CaMa Flood** — Catchment Modelling and Analysis (routing model)

- **CAMELS DK** — Catchment Attributes and MEteoroLogy for Large-sample Studies, Denmark
- **CDF** — Cumulative Distribution Function (matching technique)
- **CDI** — Copernicus Drought Index
- **CRPS** — Continuous Ranked Probability Score
- **CSE-UA** — Shuffled Complex Evolution University of Arizona (optimisation algorithm)
- **DFF** — Danish Independent Research Fund
- **DK model** — Danish hydrological model
- **DMI** — Danish Meteorological Institute
- **dPL** — Differentiable Parameter Learning
- **DREAM** — Drought Resilience through satellite-integrated Early warning And Monitoring
- **EDgE** — ECMWF Drought Ensemble Global forecasting
- **ECMWF** — European Centre for Medium-Range Weather Forecasts
- **ECMWF IFS** — ECMWF Integrated Forecasting System
- **EDO CDI v4** — European Drought Observatory Copernicus Drought Index version 4
- **EFAS Seasonal** — European Flood Awareness System Seasonal forecasts
- **ESP** — Ensemble Streamflow Prediction
- **ESA CCI** — European Space Agency Climate Change Initiative (soil moisture)
- **ESSD** — Earth System Science Data (journal)
- **FEST** — Flash-flood Event-based Spatially-distributed Terrain-following (hydrological model)
- **GEUS** — Geological Survey of Denmark and Greenland
- **GIA** — Glacial Isostatic Adjustment
- **GMD** — Geoscientific Model Development (journal)
- **GRACE FO** — Gravity Recovery and Climate Experiment Follow-On
- **HIP** — Hydrological Information Platform (Danish system)
- **HIPRAD** — HIP Radar database
- **HPC Claaudia** — High-Performance Computing facility at Aalborg University
- **ICON** — ICOsahedral Nonhydrostatic model (DWD/ECMWF)
- **ICE-6G_D** — Glacial isostatic adjustment model (version 6G with denser temporal sampling)
- **LETKF** — Local Ensemble Transform Kalman Filter
- **LSTM** — Long Short-Term Memory (neural network architecture)
- **MCDI** — Multi-pillar Combined Drought Index (new product)
- **MCMC** — Markov Chain Monte Carlo
- **MODFLOW** — Modular Three-Dimensional Finite-Difference Groundwater Flow Model
- **MSc** — Master of Science
- **NeuralHydrology** — Differentiable hydrology framework
- **PyGLDA** — Python Geodesy and Land Data Assimilation (AAU research infrastructure)
- **ROC** — Receiver Operating Characteristic
- **SEAS5** — Seasonal Forecasting System 5 (ECMWF)
- **SMAP** — Soil Moisture Active Passive (NASA satellite mission)
- **SPEI** — Standardised Precipitation-Evapotranspiration Index
- **SPI** — Standardised Precipitation Index
- **SRI** — Standardised Runoff Index
- **SSI** — Standardised Streamflow Index
- **SSMI** — Standardised Soil Moisture Index
- **SGI** — Standardised Groundwater Index
- **SIMGM** — Simulating Groundwater Management (hydrological model)
- **STOTEN** — Science of The Total Environment (journal)
- **USDM** — United States Drought Monitor (severity classification)
- **VIC** — Variable Infiltration Capacity (macroscale land surface model)
- **VICGlobal** — Global parameter set for VIC
- **W3RA** — Water Resources Reanalysis (Australian Bureau of Meteorology)
- **WGHM** — WaterGAP Global Hydrology Model
- **WP1–WP5** — Work Package 1 through 5